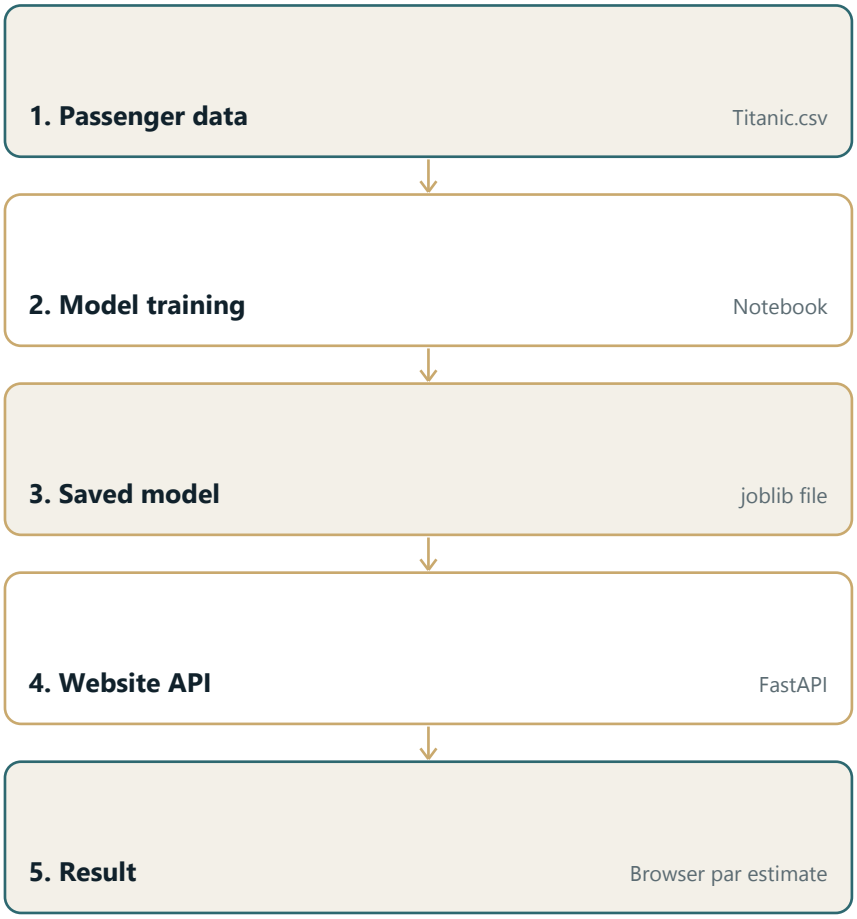
TITANIC SURVIVAL PROJECT

# Project ko 1 minute mein samjhein

Yeh website passenger ki details leti hai aur trained model se survival chance ka estimate dikhati hai. Saath mein CSV ke patterns explore kar sakte hain.

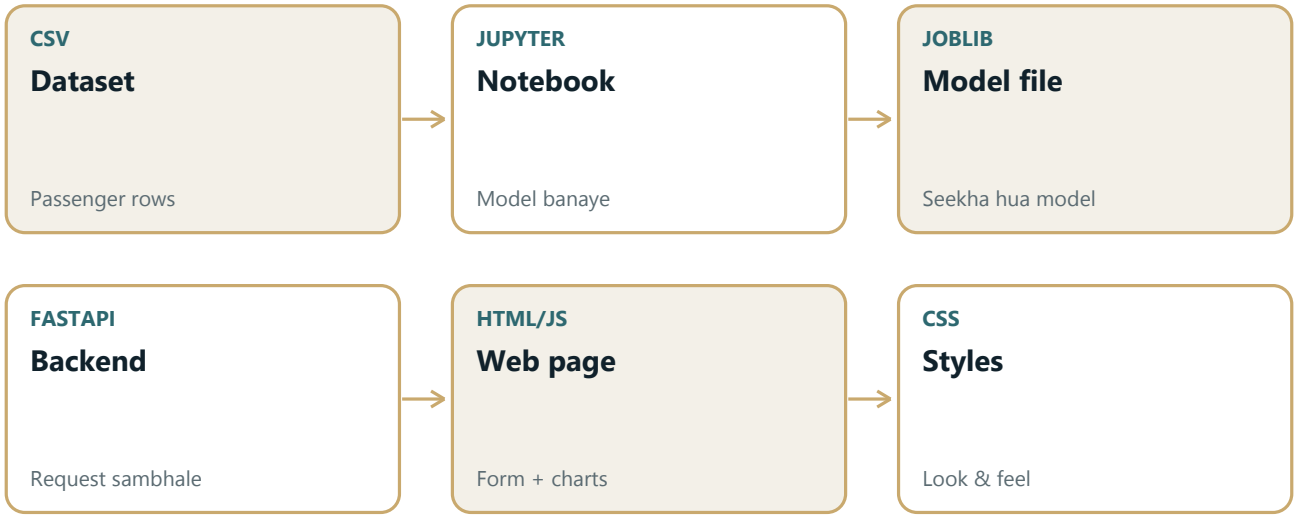



## Seedhi baat: data  model  website  estimate.

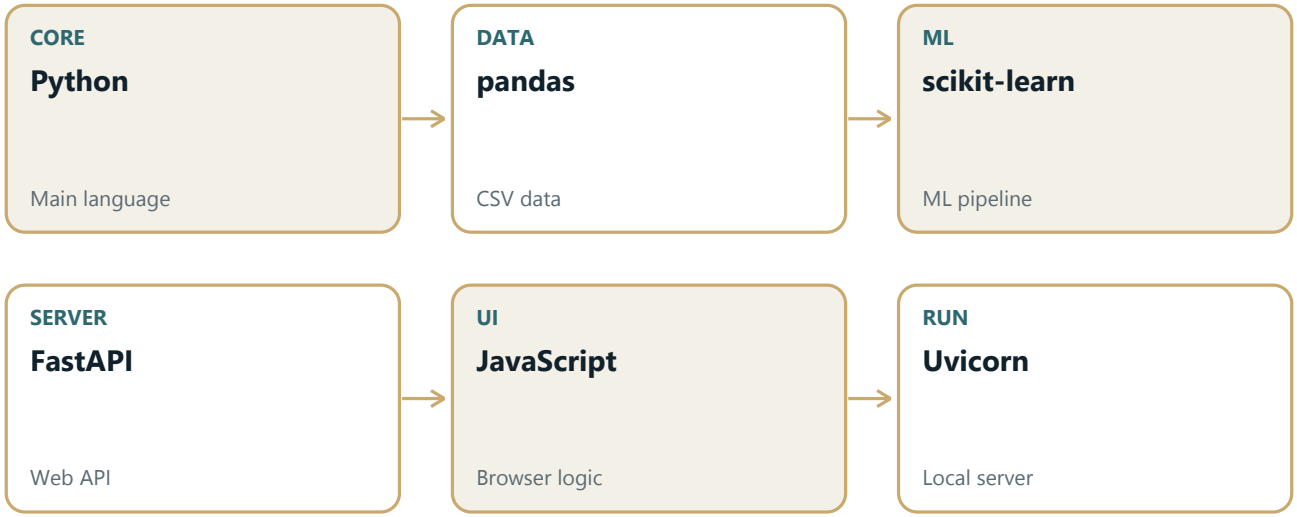
Yeh historical data par bani learning demo hai. Estimate guarantee nahi hota.

# Project ke parts

Har file ka ek simple kaam hai.



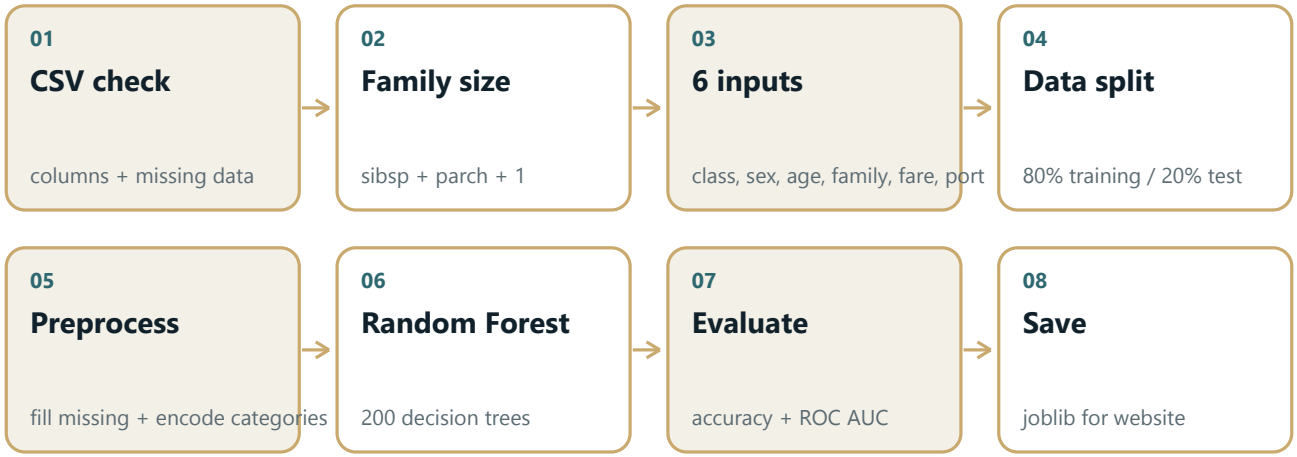

## Technology map



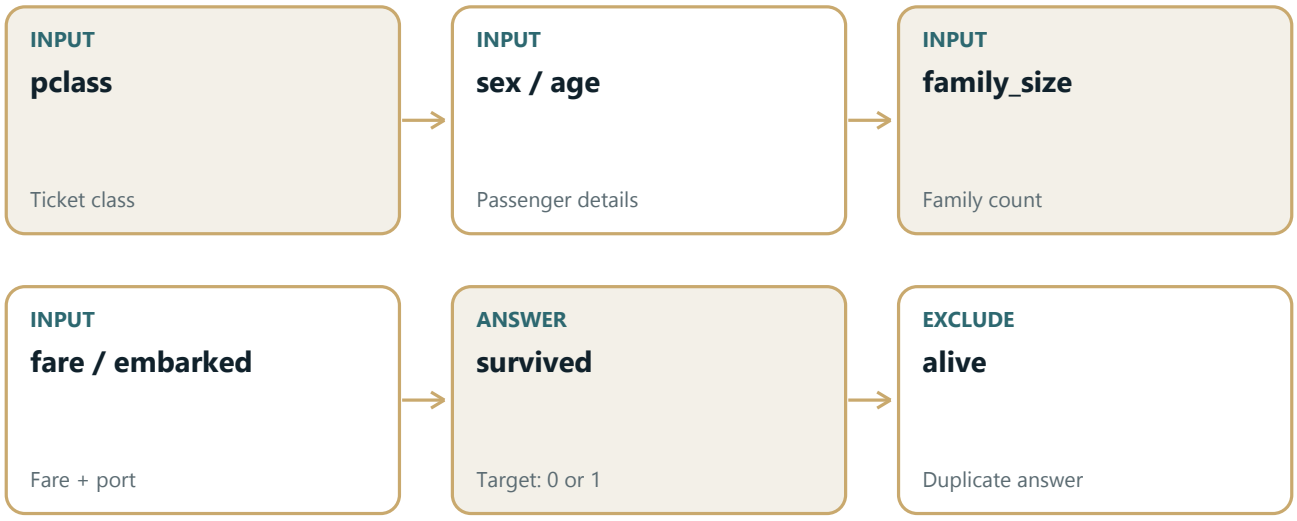

Key files: app.py · Titanic.csv · Titanic_Survival_Analysis.ipynb · titanic_survival_pipeline.joblib · static/index.html, app.js, explore.js, manifest.css

# Model kaise seekhta hai?

Notebook data ko prepare karke Random Forest train karti hai. Website wahi saved pipeline dobara use karti hai.

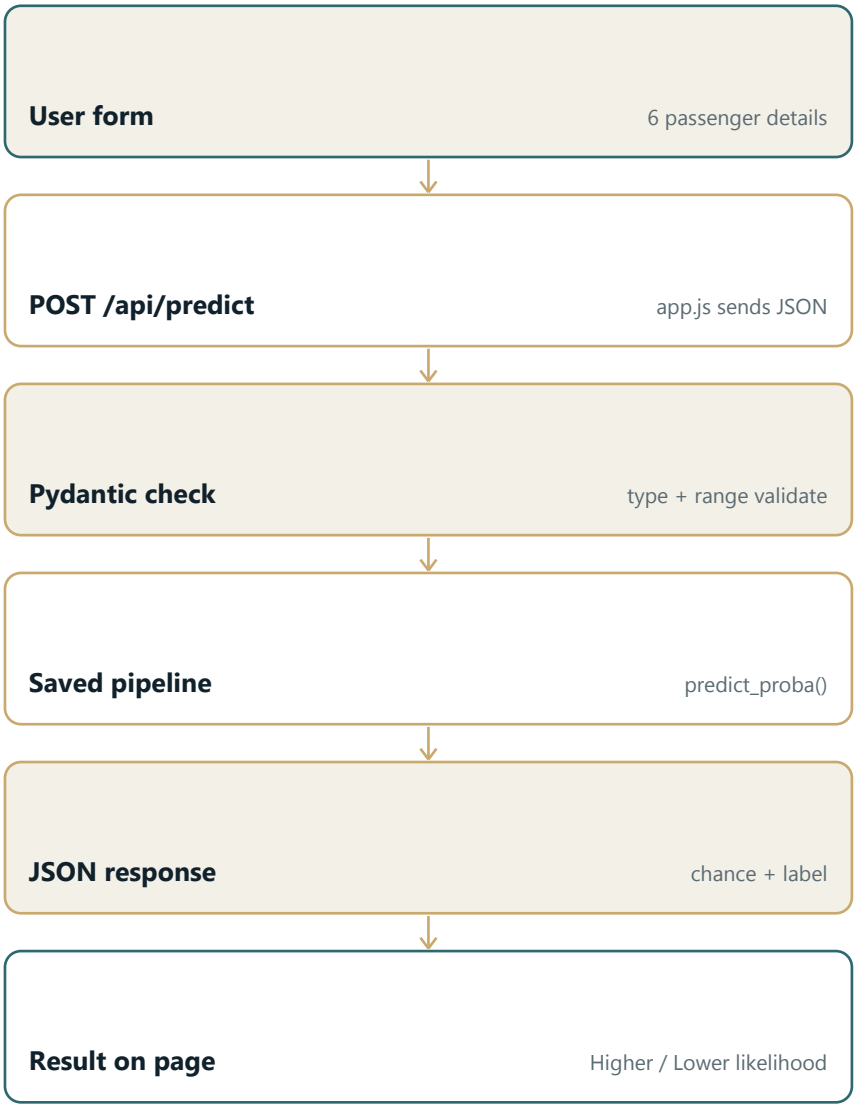



## Important fields

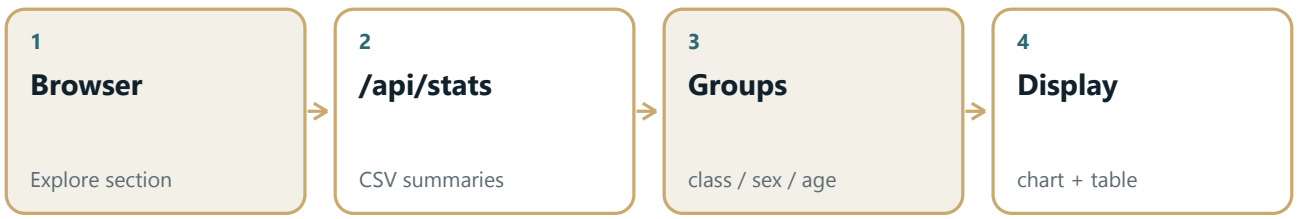


# Website request ka raasta

Form se API tak: browser details bhejta hai, backend validate karta hai, model chance return karta hai.




## Data explorer ka raasta



# Local setup aur team notes

Project ko apne computer par chalane ke liye yeh commands use karein.

## Run kaise karein?

PowerShell commands

```
python -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install -r requirements.txt
python -m uvicorn app:app --reload
Open: http://127.0.0.1:8000
```

Model missing ho? Notebook open karo  Run All  phir server restart.
Metric yaad rakhein: website par validation accuracy 79.9% dikhayi gayi hai. Demo output ek estimate hai, promise nahi.